

— Practical guide for operators

Turn Papers into a Knowledge Graph

Install to upload — with copy-paste code, real outputs, recipes, and troubleshooting. No theory you don't need.

INPUT	plain text extracted from a paper (not a PDF)
PROCESS	Stage 1 facts → Stage 2 frames → Stage 3 ontology mapping
OUTPUT	validated N-Quads — a graph you can load

```
text >> analyze_paper() >> results >> export_neptune_nquads() >> .nq
```

In this manual

Everything is task-oriented: find what you want to do, copy the code, run it.

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What it does & what you need

You give it the text of a paper. It gives you back structured, verified, source-anchored facts — and, optionally, a knowledge-graph file ready to upload.

› HOW IT WORKS, IN 30 SECONDS

Three stages run automatically for each paragraph:

STAGE	WHAT IT PRODUCES
Stage 1	Atomic facts , each labeled (measurement? cause? prediction?) and tied to its exact source text.
Stage 2	Verified frames — the facts organized into fields, grounding-checked, with a confidence score.
Stage 3	The facts mapped onto an ontology and emitted as RDF for a graph database (optional — only if you give it an ontology).

› WHAT YOU NEED BEFORE YOU START

PYTHON 3.10+

It's a library you import — not an app or a server.

A LANGUAGE MODEL

Your own — an API key (Anthropic/OpenAI) or a local/hosted model. The library brings none.

TEXT, NOT PDFS

Convert PDFs to text first (see Chapter 2). The library reads plain text.

AN ONTOLOGY (STAGE 3 ONLY)

An `.owl` / `.ttl` file for your subject area. Skip it to get just facts.

MENTAL MODEL —

You supply the electricity (your model) and, optionally, the blueprint (your ontology). The library is the factory that turns text into structured knowledge.

Install & set up

1 • Install the library

Pick the extras you need — they're additive:

COMMAND	GIVES YOU
<code>pip install graphrag-stage1</code>	Stage 1 + Stage 2 (facts & frames)
<code>pip install "graphrag-stage1[ontology]"</code>	+ Stage 3 ontology mapping
<code>pip install "graphrag-stage1[validation]"</code>	+ <code>validate()</code> output checking
<code>pip install "graphrag-stage1[anthropic]"</code>	+ built-in Claude client
<code>pip install "graphrag-stage1[openai]"</code>	+ built-in OpenAI client

Most users want everything:

```
pip install "graphrag-stage1[ontology,validation,anthropic]"
```

2 • Give it access to your model

For the built-in clients, set the standard environment variable:

```
export ANTHROPIC_API_KEY=sk-... # or OPENAI_API_KEY=sk-...
```

3 • Turn your PDF into text

The library reads text. A few lines with PyMuPDF handle most PDFs:

```
pip install pymupdf
```

```
import fitz # PyMuPDF
text = "\n\n".join(page.get_text() for page in fitz.open("paper.pdf"))
```

ON AWS —

Use **Amazon Textract** for the PDF→text step. For scientific papers with complex layout, **GROBID** gives cleaner results.

4 • (Optional) Place your ontology

For Stage 3, have your domain ontology file ready (Chapter 6). Skip this to get just facts.

Your first run

The entire program is a few lines. Here it is, then a walk-through.

```
from graphrag_stage1 import AnthropicClient, analyze_paper, validate

text = open("paper.txt", encoding="utf-8").read()

results = analyze_paper(
    text,
    client=AnthropicClient(),          # your model
    max_concurrency=16,                # speed dial
    domain_ontology="my_domain.owl",   # omit this line to skip Stage 3
)

for r in results:                     # one entry per paragraph, in order
    if "error" in r:                  # a failed paragraph never stops the run
        continue
    for fact in r["stage1"]["statements"]:
        print(fact["facets"]["relation"], "|", fact["text"])
```

> WHAT EACH LINE DOES

- `AnthropicClient()` — the model. Reads `ANTHROPIC_API_KEY`. Swap for `OpenAIClient()`, `OllamaClient()`, or your own (Chapter 5).
- `analyze_paper(text, ...)` — splits the text into paragraphs and runs all stages in parallel.
- `max_concurrency=16` — how many model calls run at once (Chapter 8).
- `domain_ontology=...` — loads the core ontologies plus your domain file and runs Stage 3. Leave it out for Stage 1 + 2 only.
- `results` — a list, one item per paragraph, in the same order as the paper.

WHAT YOU'LL SEE —

Lines like `CAUSATION | The fused sensor data reduced positioning error by 35 percent.` — each extracted fact and its relationship type.

Reading the results

`results` is a list — one entry per paragraph. Each entry has `stage1`, `stage2`, and (if you passed an ontology) `stage3`.

› THE MOST USEFUL PART: THE FACTS

```
fact = results[0]["stage1"]["statements"][0]

fact["text"]          # "The fused sensor data reduced positioning error by 35 percent."
fact["facets"]["relation"] # "CAUSATION"
fact["facets"]["modality"] # "OBSERVED" (fact vs. prediction vs. hypothesis)
fact["facets"]["polarity"] # "POSITIVE" (or "NEGATED")
fact["provenance"]     # where it came from: char span, page, verbatim source text
fact["confidence"]["overall"] # a number (see the note in Chapter 10)
```

› WHAT EVERY FIELD TELLS YOU

YOU WANT...	READ THIS
The fact, in plain words	<code>fact["text"]</code>
Its labels (cause, measurement, prediction...)	<code>fact["facets"]</code>
Where it came from (the citation)	<code>fact["provenance"]</code>
Reasoning links between facts	<code>results[i]["stage1"]["relations"]</code>
The verified, field-by-field version	<code>results[i]["stage2"]["frames"]</code>
The knowledge-graph mapping	<code>results[i]["stage3"]</code>

› CHECK THE OUTPUT IS WELL-FORMED

```
from graphrag_stage1 import validate
validate(results[0]["stage1"]) # returns it, or raises if the shape is wrong
validate(results[0]["stage2"])
```

FAILED PARAGRAPHS –

If a paragraph couldn't be processed, its entry is `{"error": "...", "paragraph_id": "..."} instead — check for "error" before reading "stage1". One failure never stops the rest of the paper.`

Choose your model

The library works with any model that can return JSON. Built-ins cover the common cases with zero code; anything else is a tiny adapter.

› BUILT-IN (NO CODE)

```
from graphrag_stage1 import AnthropicClient, OpenAIClient

AnthropicClient()                # default Claude, reads ANTHROPIC_API_KEY
AnthropicClient(model="claude-sonnet-5")  # pick any model
OpenAIClient(model="gpt-4o-mini")        # reads OPENAI_API_KEY
```

› ANY OPENAI-COMPATIBLE ENDPOINT (VLLM, GROQ, TOGETHER, LOCAL SERVERS)

```
import openai
from graphrag_stage1 import OpenAIClient
client = OpenAIClient(sdk=openai.OpenAI(base_url="http://my-server/v1", api_key="..."))
```

› LOCAL, OFFLINE (OLLAMA)

```
from graphrag_stage1 import OllamaClient
client = OllamaClient(model="qwen3:8b")  # talks to a local Ollama server
```

› YOUR OWN MODEL — IMPLEMENT ONE METHOD

```
class MyClient:
    model_id = "my-provider/my-model"  # recorded in every result's provenance
    def complete(self, prompt, schema, *, stronger=False):
        # call your model; force JSON matching `schema`; return the dict
        ...
```

TWO THINGS TO KNOW —

Speed & quality depend on the model you choose — a fast model (Haiku, gpt-4o-mini) for throughput, a stronger one for accuracy. And your `complete` is called from many threads at once, so it must be **thread-safe** (the built-ins already are).

SEE ALSO —

A full runnable cookbook for every provider — Claude, GPT, Azure, Bedrock, Gemini, vLLM, Ollama — with pros/cons is in `examples/model_clients.py` and `docs/CHOOSING_A_MODEL.md`.

Use your own ontology

Three foundational ontologies (BFO, IAO, CCO) are always loaded. You add one domain ontology — the one that matches your papers.

› WHERE FILES LIVE

```
ontologies/
├─ bfo.owl           always loaded (required)
├─ iao.owl           always loaded (required)
├─ CommonCoreOntologies*.ttl  always loaded (required)
├─ Relations/ Alignment/  supporting files (leave as-is)
└─ Domain/<your_ontology>.owl ← YOUR file goes here
```

› TWO WAYS TO PLUG IN YOUR DOMAIN ONTOLOGY

```
# Option A – pass the path directly (easiest):
results = analyze_paper(text, client=c, domain_ontology="/path/to/my_domain.owl")

# Option B – drop the file into ontologies/Domain/ and point at it:
results = analyze_paper(text, client=c, domain_ontology="ontologies/Domain/my_domain.owl")
```

You can also point at a *folder* of ontology files, not just one.

› WHAT MAKES A GOOD DOMAIN ONTOLOGY

- A proper OWL/RDF file: `.owl`, `.ttl`, `.rdf`, `.nt`, `.n3`, or `.jsonld`.
- **Human-readable labels** on its classes and properties — that's what your facts get matched against.
- Ideally built on BFO (like CCO), so it stacks cleanly onto the core.

PLUG-AND-PLAY —

There are no hardcoded domain terms in the code — it reads the labels inside whatever ontology you load. Swap the file, and the mapping targets change with it. Terms it can't map confidently are logged, never guessed.

Produce the Neptune file

Stage 3 gives you the mapping. One more call turns it into the N-Quads file you upload to a graph database.

```
from graphrag_stage1.stage3_neptune_export import export_neptune_nquads

meta = export_neptune_nquads(
    results[0]["stage3"],
    destination="paragraph0.nq",          # ← the file to upload
    quarantine_destination="rejected.ttl", # optional: what was filtered out
    report_destination="report.txt",      # optional: the validation report
)
print(meta["validated_triple_count"], "triples ->", meta["destination"])
```

› WHAT HAPPENS IN THAT CALL

- The mapped facts are turned into RDF triples, each document in its own *named graph*.
- A **SHACL validation** gate checks every triple. Clean ones go to the `.nq` file; bad ones are *quarantined* to a separate file and never uploaded.
- You get back a summary: how many triples passed, how many were quarantined.

› THEN LOAD IT

```
Upload *.nq → S3 → run the Neptune bulk loader
```

NEPTUNE DOES NOT REASON FOR YOU –

Loading the file into Neptune stores and lets you query the facts — it does **not** automatically infer new ones from the ontology. OWL reasoning (deriving implied facts) is a separate step you add afterward if you want it. It is not part of this library.

Make it fast

Which dial matters depends entirely on whether your model endpoint can scale.

› ON A HOSTED API: TURN UP THE CONCURRENCY

Paragraphs are independent, so the library runs them in parallel. On an endpoint that scales, more concurrent calls really do finish sooner:

$$time \approx (number\ of\ model\ calls \times latency\ per\ call) \div max_concurrency$$

```
analyze_paper(text, client=c, max_concurrency=16, # total calls at once
              stage2_concurrency=6)             # per-paragraph fan-out
```

Set `max_concurrency` to your provider's safe concurrent-request budget. Too low is slow; too high gets you rate-limited. A ~50-paragraph paper lands in **about a minute**.

› ON A SINGLE LOCAL GPU: THAT FORMULA STOPS WORKING

Raising `max_concurrency` past a few does nothing. The GPU is not queueing your calls politely — it is already **saturated at 97–98% utilisation**, and extra concurrent requests simply time-slice the same silicon. You must make the model do *less work*, not give it more work at once.

Where the time actually goes, measured on one paragraph (qwen3, laptop RTX 5080):

CALL	READING THE PROMPT	WRITING THE ANSWER
Stage 1 (decompose)	0.1 s — 749 tokens	2.5 s — 117 tokens
Stage 2 (one frame)	0.5 s — 2,184 tokens	18.7 s — 819 tokens

Reading is nearly free; writing is the entire cost. A Stage 2 frame writes about seven times as many tokens as a Stage 1 call, and Stage 2 runs once per fact rather than once per paragraph. That is your bottleneck.

› THE THREE LEVERS THAT ACTUALLY WORK

LEVER	HOW	EFFECT
A smaller model for Stage 2	<code>stage2_client=OllamaClient(model="qwen3:4b")</code>	Stage 2 is form-filling, not deep reasoning — it doesn't need your best model.
Skip the model where the answer is already known	<code>STAGE2_HYBRID=1</code>	Measurements, classifications, mechanisms and methods get built from Stage 1's own output — no call at all.
Drop the retry pass	<code>STAGE2_MAX_RETRIES=0</code>	Each reprocessed fact is a second full call. Costs you the auto-repair pass.

DON'T BOTHER —

Shrinking or batching prompts. Reading the prompt is only ~3% of a Stage 2 call, so there is nothing there to win. And if you're on Ollama, set `OLLAMA_MAX_LOADED_MODELS=2` so an 8B Stage 1 and a 4B Stage 2 stay in memory together instead of being swapped in and out on every paragraph.

Recipes & cheat sheet

Common tasks, ready to copy.

› GET JUST THE FACTS AS PLAIN TEXT

```
facts = [s["text"] for r in results if "stage1" in r
        for s in r["stage1"]["statements"]]
```

› KEEP ONLY CAUSAL FACTS

```
causal = [s for r in results if "stage1" in r
          for s in r["stage1"]["statements"]
          if s["facets"]["relation"] in ("CAUSATION", "MECHANISM")]
```

› KEEP ONLY HIGH-CONFIDENCE FACTS

```
strong = [s for r in results if "stage1" in r
          for s in r["stage1"]["statements"]
          if s["confidence"]["overall"] >= 0.8]
```

› PROCESS A WHOLE FOLDER, STREAMING PROGRESS

```
import glob
from graphrag_stage1 import run_paper, split_paragraphs

def on_done(i, r):
    print("finished paragraph", i)

for path in glob.glob("papers/*.txt"):
    paras = split_paragraphs(open(path, encoding="utf-8").read())
    results = run_paper(paras, client=c, max_concurrency=16, on_result=on_done)
```

› THE FULL QUICK-REFERENCE

TO GET...	CALL / READ
Facts + frames (+ graph) for a paper	<code>analyze_paper(text, client=c[, domain_ontology=...])</code>
Facts for one paragraph	<code>process_paragraph(text, client=c)</code>
The Neptune file	<code>export_neptune_nquads(r["stage3"], "out.nq")</code>
The facts	<code>r["stage1"]["statements"]</code>
The reasoning links	<code>r["stage1"]["relations"]</code>
Validate an output	<code>validate(r["stage1"])</code>

Troubleshooting & FAQ

› "NO MODULE NAMED GRAPHRAG_STAGE1"

It isn't installed in the environment you're running. Run `pip install ...` in the same Python/venv, or install it into your container image.

› "IT WON'T TAKE MY PDF"

Correct — it reads **text**, not PDFs. Convert first with PyMuPDF/Texttract/GROBID (Chapter 2), then pass the text.

› STAGE 3 GIVES AN EMPTY OR THIN GRAPH

Usually a **domain-ontology mismatch** — the ontology doesn't cover your paper's subject, so terms fail closed (by design). Use an ontology that matches your field, with good labels.

› RATE-LIMIT / THROTTLING ERRORS

Lower `max_concurrency` to your provider's limit. That single cap controls total calls in flight.

› IT'S SLOW

You're likely on a single local GPU (which serializes) or `max_concurrency` is low. Use a hosted/scalable endpoint and raise the cap (Chapter 8).

› STRUCTURED-OUTPUT / JSON ERRORS FROM MY CUSTOM MODEL

Your `complete` must return a dict matching the given `schema`. Use your provider's tool-calling / JSON-schema mode (the built-in clients show the pattern). Very small local models sometimes can't do reliable structured output.

ABOUT THE CONFIDENCE SCORES —

Trust the **structure, the exact numbers, and the provenance** today — those are deterministic guarantees. The numeric **confidence values are not yet calibrated** (the evaluation set is still small), so don't gate important decisions purely on them.

WHERE TO GO NEXT —

`DOCUMENTATION.md` — the complete field-by-field API reference. `INTEGRATION.md` — integrating into another system. `DESIGN.md` — how and why it works inside.

Jupyter Notebook Workflow

A complete, runnable notebook is at `examples/jupyter_usage_demo.ipynb`. Open it, run the cells, and adapt to your papers.

› OPEN IT

```
jupyter lab examples/jupyter_usage_demo.ipynb
# or
jupyter notebook examples/jupyter_usage_demo.ipynb
```

› WHAT'S INSIDE (9 SECTIONS)

SECTION	COVERS
1. Installation	<code>%pip install graphrag-stage1</code> + optional extras (<code>[ontology]</code> , <code>[anthropic]</code> , <code>[openai]</code> , <code>[dev]</code>)
2. Imports & LLM Client	All imports; choose Anthropic, OpenAI, Ollama, or custom <code>LLMClient</code>
3. Stage 1 Only	<code>process_paragraph()</code> → Knowledge Artifact Graph (propositions, discourse relations, coreference)
4. Stage 1 → 2	<code>stage2_pipeline()</code> → Semantic frames with typed slots, grounding checks, confidence
5. Stage 3 (Ontology)	<code>OntologyManager</code> + <code>run_pipeline(ontology=...)</code> → RDF/JSON-LD + Turtle export
6. Full Paper	<code>analyze_paper()</code> / <code>run_paper()</code> for multi-paragraph parallel processing
7. Validation	<code>validate()</code> against versioned JSON Schemas
8. Exploration & Debugging	Inspect provenance, mapping audits, enable debug logging
9. Patterns & Tips	Custom clients, checkpointing, error handling, performance tuning, output formats

› MINIMAL RUNNABLE EXAMPLE (3 CELLS)

```
# Cell 1: Install
%pip install graphrag-stage1

# Cell 2: Imports + client
from graphrag_stage1 import run_pipeline, OllamaClient
client = OllamaClient() # or AnthropicClient(), OpenAIClient()

# Cell 3: Run pipeline
text = "The fused sensor data reduced positioning error by 42% because it provides a more complete s
result = run_pipeline(text, client=client, paragraph_id="demo:p1")

# Cell 4: Inspect outputs
kag = result["stage1"]      # Knowledge Artifact Graph
frames = result["stage2"]   # Validated semantic frames

print(f"Propositions: {len(kag['propositions'])}")
print(f"Frames: {len(frames['frames'])}")
for f in frames["frames"]:
    print(f"  {f['frame_type']}: confidence={f['confidence']:.2f}")
```

› WITH STAGE 3 (ONTOLOGY MAPPING)

```
# Requires: %pip install "graphrag-stage1[ontology]"
# And: ontologies/ directory with core + your domain ontology

from graphrag_stage1.ontology_manager import OntologyManager

ontology = OntologyManager("ontologies").load_all(domain_ontology="Domain/your_domain.owl")

result = run_pipeline(
    text, client=client, paragraph_id="demo:p1",
    ontology=ontology # <-- triggers Stage 3
)

stage3 = result["stage3"]
print(f"Mapped frames: {len(stage3['mapped_frames'])}")
print(f"Ontology individuals: {len(stage3['ontology_individuals'])}")

# Export for Neptune / GraphDB
from graphrag_stage1.stage3_jsonld import build_jsonld
jsonld = build_jsonld(stage3)
import json
with open("output.jsonld", "w") as f:
    json.dump(jsonld, f, indent=2)
```

› KEY PATTERNS FOR NOTEBOOK WORKFLOWS

PATTERN	CODE SNIPPET
Checkpointing	<code>run_paper(paras, client=c, on_result=lambda i, r: save_checkpoint(i, r))</code>
Error handling	<code>if "error" in r: log(r["error"]); continue</code>
Provenance citation	<code>prop["provenance"]["source_statement"]</code> gives verbatim text span
Debug LLM calls	<code>logging.getLogger("graphrag_stage1.llm").setLevel(logging.DEBUG)</code>
Batch + aggregate	Collect all <code>stage3["mapped_frames"]</code> → merge → single KG

> ENVIRONMENT VARIABLES FOR LOCAL DEVELOPMENT

```
# .env or shell export
export STAGE1_MODEL="qwen3:8b"
export STAGE2_MODEL="qwen3:8b"
export STAGE2_STRONGER_MODEL="qwen3:14b"
export OLLAMA_URL="http://localhost:11434/api/generate"
```

```
# Or set in notebook:
import os
os.environ["STAGE1_MODEL"] = "qwen3:8b"
os.environ["STAGE2_MODEL"] = "qwen3:8b"
```

FULL REFERENCE –

[DOCUMENTATION.md](#) §21 has the complete notebook guide with all sections.